

Bloom's triplet weighted-BMO condition is not necessary in dimension three

Counterexample packet for arXiv:2301.00617

11 August 2026

Status. Complete counterexample to higher-dimensional necessity of Bloom's triplet condition, likely valid pending human review. Theorem-level overlap with a relevant 2026 paper has not been excluded, so no priority claim is made.

1 Source question and precise scope

In Remark 7.11 of [3], Hytönen considers a matrix weight

$$W = U^* \Lambda U, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n),$$

where U is unitary and every λ_k is a scalar A_2 weight. He asks what conditions on U ensure that W is a matrix A_2 weight. Bloom's triplet condition, in the convention of [4, Definition 2.1 and Theorem 2.2], is

$$u_{ij} \in \text{BMO}_{\sqrt{\lambda_i/\lambda_k}} \quad \text{for every } i, j, k. \quad (1)$$

It is sufficient, and Bloom proved it necessary for 2×2 matrices [1, 2]. The source says the general case appears open.

7.11. *Remark.* Aside from the examples already discussed, the generalised commutators $\vec{a} \cdot T \vec{b}$ also arise in the following question studied by Bloom [4, 5]. Suppose that a matrix weight W is given in the diagonalised form $W = U^* \Lambda U$, where U is unitary, Λ is diagonal, and the diagonal entries λ_k of Λ are scalar A_2 weights. What does one need to know about U in order to conclude that $W \in A_2$? (According to [5, Theorem 4.2], the condition that $\lambda_k \in A_2$ is necessary for $W \in A_2$, if in addition U is assumed to be continuous.)

Let T be the Hilbert transform, or another operator whose boundedness on the matrix-weighted $L^2(W)$ characterises $W \in A_2$. By connecting the $L^2(W)$ boundedness of T to the boundedness of the classical commutators $[T, \bar{u}_{ij}]$ between the weighted spaces $L^2(\lambda_i)$ and $L^2(\lambda_k)$ (sic: the condition involves triplets of indices (i, j, k)), [4, Theorem 5.1] shows that $u_{ij} \in \text{BMO}_{\sqrt{\lambda_i/\lambda_k}}$ (a weighted BMO space, nowadays commonly referred to as Bloom-type BMO) is a sufficient condition. In the special case of 2×2 matrices, it is also necessary by [5, Theorem 4.3] but, over 30 years since these contributions, the general case seems to remain open. (The author is grateful to Amalia Culiuc for bringing this question to his attention [9].)

The theorem below disproves necessity of (1) in dimension three. It does not give a complete classification of all admissible U .

2 Definitions and proof intuition

For an interval I , write $\langle F \rangle_I = |I|^{-1} \int_I F$. We use the squared matrix characteristic

$$[W]_{A_2} := \sup_I \left\| \langle W \rangle_I^{1/2} \langle W^{-1} \rangle_I^{1/2} \right\|^2.$$

For a scalar weight ν , Bloom's weighted BMO seminorm here is

$$\|b\|_{\text{BMO}_\nu} := \sup_I \frac{1}{\nu(I)} \int_I |b - b_I| dx, \quad b_I = |I|^{-1} \int_I b, \quad \nu(I) = \int_I \nu.$$

Proof intuition. The entrywise test (1) compares every entry of U against every eigenvalue scale. Matrix A_2 , however, cannot see rotations inside an eigenvalue cluster whose members are pointwise comparable. We rotate a two-dimensional cluster with eigenweights s^{-1} and $2s^{-1}$ as rapidly as desired. This preserves matrix A_2 by pointwise comparison with a block-scalar diagonal weight. A third eigenweight $4s$ makes the Bloom denominator for one rotating entry tend to zero at infinity, while the entry continues to oscillate by a fixed amount on every period.

3 Two elementary lemmas

Lemma 1 (Loewner comparability). *Let D, W be positive-definite matrix weights and suppose $cD \leq W \leq CD$ almost everywhere, where $0 < c \leq C < \infty$. If $D \in A_2$, then $W \in A_2$, and*

$$[W]_{A_2} \leq \frac{C}{c} [D]_{A_2}.$$

Proof. Inversion reverses Loewner order, so $C^{-1}D^{-1} \leq W^{-1} \leq c^{-1}D^{-1}$. Fix I and put $A = \langle W \rangle_I$, $B = \langle W^{-1} \rangle_I$, $A_0 = \langle D \rangle_I$, and $B_0 = \langle D^{-1} \rangle_I$. Then $A \leq CA_0$ and $B \leq c^{-1}B_0$. Hence

$$\begin{aligned} \|A^{1/2}B^{1/2}\|^2 &= \|B^{1/2}AB^{1/2}\| \\ &\leq C\|B^{1/2}A_0B^{1/2}\| = C\|A_0^{1/2}BA_0^{1/2}\| \\ &\leq \frac{C}{c}\|A_0^{1/2}B_0A_0^{1/2}\| = \frac{C}{c}\|A_0^{1/2}B_0^{1/2}\|^2. \end{aligned}$$

Taking the supremum proves the claim. $\square$

Lemma 2. *For every $|\gamma| < 1$, the scalar weight $x \mapsto (1 + x^2)^{\gamma/2}$ belongs to $A_2(\mathbb{R})$.*

Proof. It is comparable to $(1 + |x|)^\gamma$. If an interval is farther from the origin than its length, this latter weight is comparable to a constant on the interval. Otherwise the interval is contained in a fixed dilation of $[-|I|, |I|]$, and direct integration of $(1 + t)^{\pm\gamma}$ gives

$$\langle (1 + |x|)^\gamma \rangle_I \langle (1 + |x|)^{-\gamma} \rangle_I \lesssim_\gamma 1,$$

because both exponents are greater than -1 . Scalar A_2 is preserved by pointwise comparability. $\square$

4 Analytic simple-spectrum counterexample

Theorem 1. *For every $a \in (0, 1)$ there exist globally real-analytic functions $U : \mathbb{R} \rightarrow SO(3)$ and $0 < \lambda_1 < \lambda_2 < \lambda_3$ such that every λ_i is scalar A_2 , the matrix weight $W = U^* \text{diag}(\lambda_1, \lambda_2, \lambda_3)U$ belongs to matrix A_2 , but Bloom's condition (1) fails. In particular, that condition is not necessary in dimension three.*

Proof. Let

$$s(x) = (1 + x^2)^{a/2}, \quad (\lambda_1, \lambda_2, \lambda_3) = (s^{-1}, 2s^{-1}, 4s),$$

and let

$$U(x) = \begin{pmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

These functions are real analytic. Since $s \geq 1$, $\lambda_1 < \lambda_2 < \lambda_3$ everywhere. Lemma 2 shows that all three eigenweights are scalar A_2 .

Set

$$D(x) = \text{diag}(s^{-1}, s^{-1}, 4s).$$

The first block of $W = U^*AU$ is an orthogonal conjugate of $\text{diag}(s^{-1}, 2s^{-1})$, while the last coordinate is $4s$. Therefore

$$D(x) \leq W(x) \leq 2D(x) \quad (x \in \mathbb{R}).$$

The diagonal weight D is matrix A_2 : its characteristic is the maximum of the scalar characteristics of s^{-1} and $4s$. Lemma 1 now gives $W \in A_2$.

It remains to disprove (1). Choose $(i, j, k) = (1, 1, 3)$. Then $u_{11}(x) = \cos x$ and

$$\sqrt{\frac{\lambda_1(x)}{\lambda_3(x)}} = \frac{1}{2s(x)} =: \nu(x).$$

For $N \geq 1$, let $I_N = [2\pi N, 2\pi(N+1)]$. Since $(\cos)_{I_N} = 0$,

$$\int_{I_N} |\cos x - (\cos)_{I_N}| dx = 4.$$

On the other hand,

$$\nu(I_N) = \frac{1}{2} \int_{I_N} (1+x^2)^{-a/2} dx \leq \pi(1+(2\pi N)^2)^{-a/2} \rightarrow 0.$$

Thus the weighted-BMO quotients are at least $4/\nu(I_N) \rightarrow \infty$. Consequently $u_{11} \notin \text{BMO}_{\sqrt{\lambda_1/\lambda_3}}$, although $W \in A_2$. $\square$

Corollary 1. *The same conclusion, still with globally simple spectrum, holds in every dimension $n \geq 3$: append distinct eigenweights $c_4s, \dots, c_ns$ with $4 < c_4 < \dots < c_n$ and extend U by the identity.*

5 The structural mechanism

Theorem 2 (Comparable-eigenvalue blocks). *Partition $\{1, \dots, n\}$ into blocks $B_1, \dots, B_m$. Suppose that for each r there is a scalar A_2 weight w_r and constants $0 < c_r \leq C_r < \infty$ such that*

$$c_rw_r \leq \lambda_i \leq C_rw_r \quad (i \in B_r).$$

If $U(x)$ is any measurable block-diagonal unitary for this partition, then $U^ \text{diag}(\lambda_i)U$ is a matrix A_2 weight. No smoothness or oscillation condition on U inside a block is needed.*

Proof. Let $D_0 = \text{diag}(w_{r(i)})$, which is scalar on each block and hence commutes with U . With $c = \min c_r$ and $C = \max C_r$,

$$cD_0 = U^*(cD_0)U \leq U^* \text{diag}(\lambda_i)U \leq U^*(CD_0)U = CD_0.$$

The diagonal weight D_0 is matrix A_2 , so Lemma 1 applies. $\square$

This theorem explains exactly why the triplet condition is too strong: it tests a within-cluster rotation against eigenweights outside the cluster even though matrix A_2 is already guaranteed by block comparability.

6 Verification, limitations, and novelty audit

The accompanying script checks positivity, the pointwise generalized eigenvalue bounds $D \leq W \leq 2D$, strict ordering of the eigenvalues, and the growth of the explicit weighted-BMO lower bound for several a and N . These computations are sanity checks; the displayed argument is the proof.

Eight focused upgrade attempts were recorded in the accompanying attempt note. They removed repeated eigenvalues, upgraded the construction to global analyticity, produced a separate local Hölder example, generalized the mechanism to arbitrary comparable blocks, and tested possible repaired necessity statements. The broad classification problem for arbitrary U remains open.

A bounded search on 11 August 2026 covered the run indexes; Bloom’s 1985 and 1989 papers; Nielsen–Šikić’s 2021 extension [4]; work on matrix power weights; the authors’ 2025 general-bases paper; and searches by the exact triplet formula, source id, title, authors, and comparable-eigenvalue phrases. No exact construction was located. However, the March 2026 paper [5] explicitly concerns diagonalisation, weighted BMO, and structural differences between compact and noncompact domains. Its primary full text could not be checked because the publisher route presented a CAPTCHA. Exact overlap is therefore unresolved and this packet makes no novelty or priority claim.

Human-review recommendation. Verify Lemma 1 and the one-period calculation in Theorem 1. If they pass, the counterexample is mathematically complete. Before assigning novelty, compare the exact construction and Theorem 2 with every theorem and example in [5].

References

- [1] S. Bloom, *A commutator theorem and weighted BMO*, Trans. Amer. Math. Soc. 292(1), 103–122 (1985).
- [2] S. Bloom, *Applications of commutator theory to weighted BMO and matrix analogs of A_2* , Illinois J. Math. 33(3), 464–487 (1989).
- [3] T. P. Hytönen, *Some remarks on convex body domination*, arXiv:2301.00617 (2023), Remark 7.11.
- [4] M. Nielsen and H. Šikić, *Muckenhoupt Matrix Weights*, J. Geom. Anal. 31, 8850–8865 (2021), doi:10.1007/s12220-020-00440-z.
- [5] M. Nielsen and H. Šikić, *Matrix weights on compact and non-compact domains*, J. Math. Anal. Appl. 555(2), 130069 (2026), doi:10.1016/j.jmaa.2025.130069.